37th
INTERNATIONAL CAE
CONFERENCE
AND EXHIBITION

VICENZA, ITALY
NOVEMBER 17 - 19
2021

HYBRID EVENT

# Flexible stand-up pouch filling process analysis by Coupling Particle-Based CFD and Multi-Flexible Body Dynamics Simulation

Matteo Berlato (University of Trento), Davide Marini (EnginSoft)

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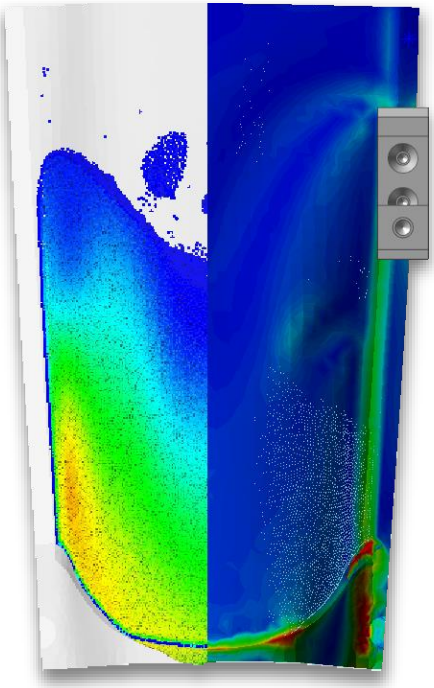


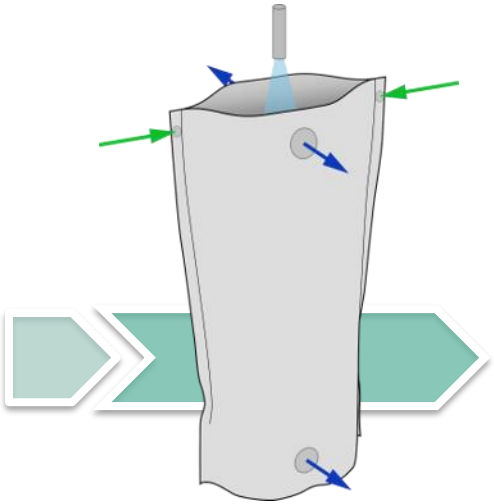

- Case study, objectives, method
- Modeling
- Simulations and results
- Conclusions and future work

# Case study

- Horizontal stand-up pouching automatic machine

# Objectives

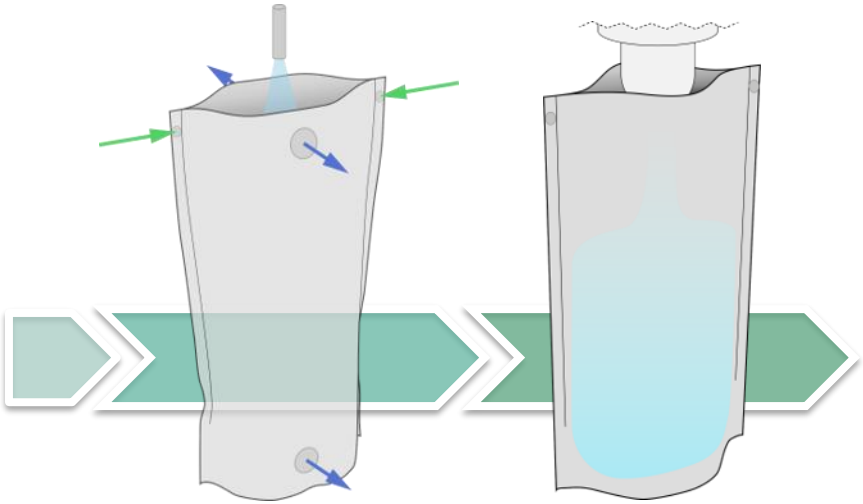
- Evaluate the influence of material properties and machine setup on the stability of the open pouch



Opening

# Objectives

- Simulate the filling process with different fluids

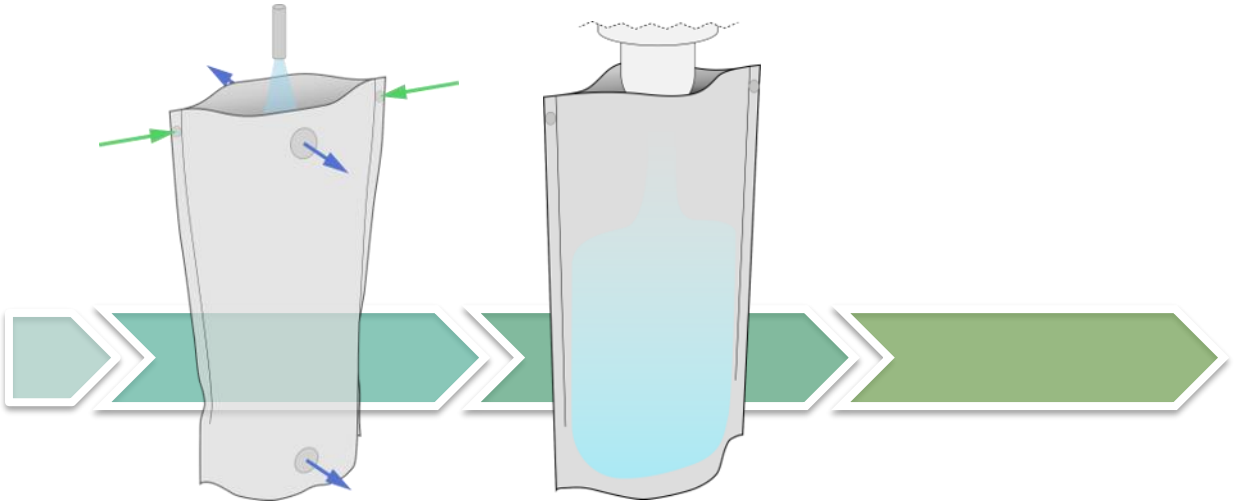


Opening

Filling

# Objectives

- Study the fluid sloshing during transportation

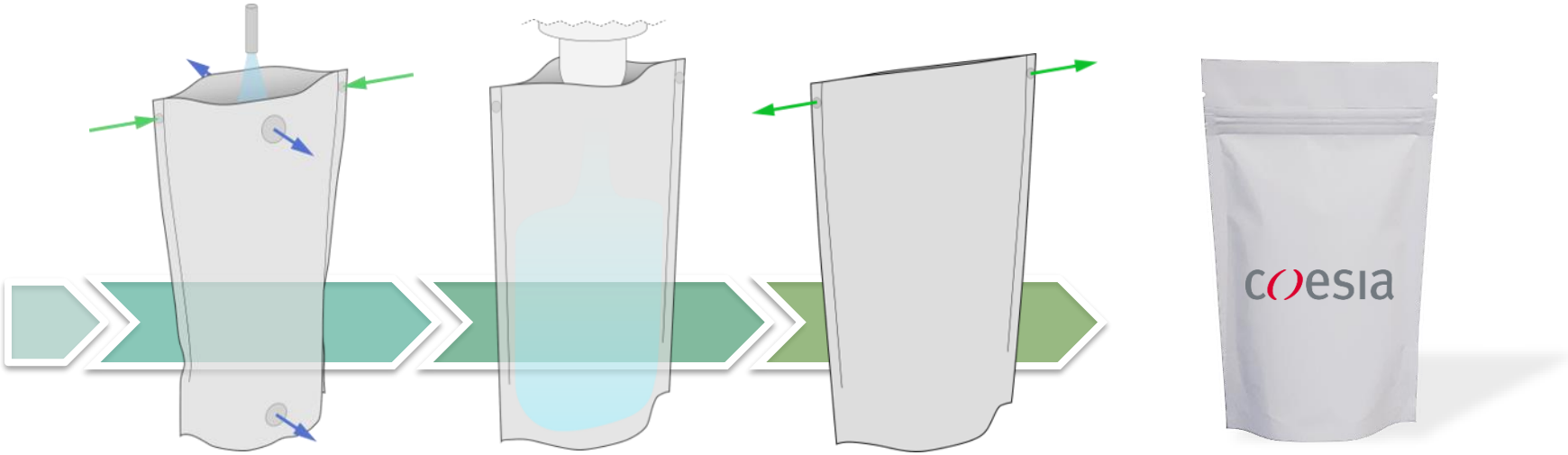


Opening

Filling

# Objectives

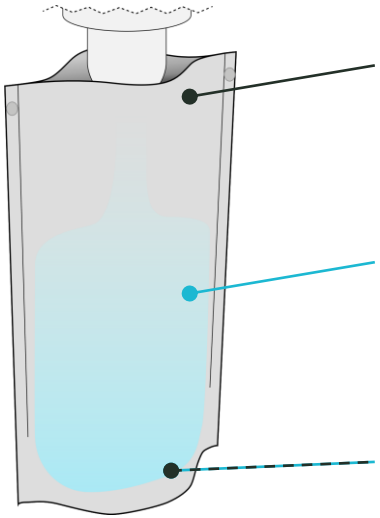
- Reproduce the pouch configuration before sealing




Opening Filling Closing

# Method

- Features and properties



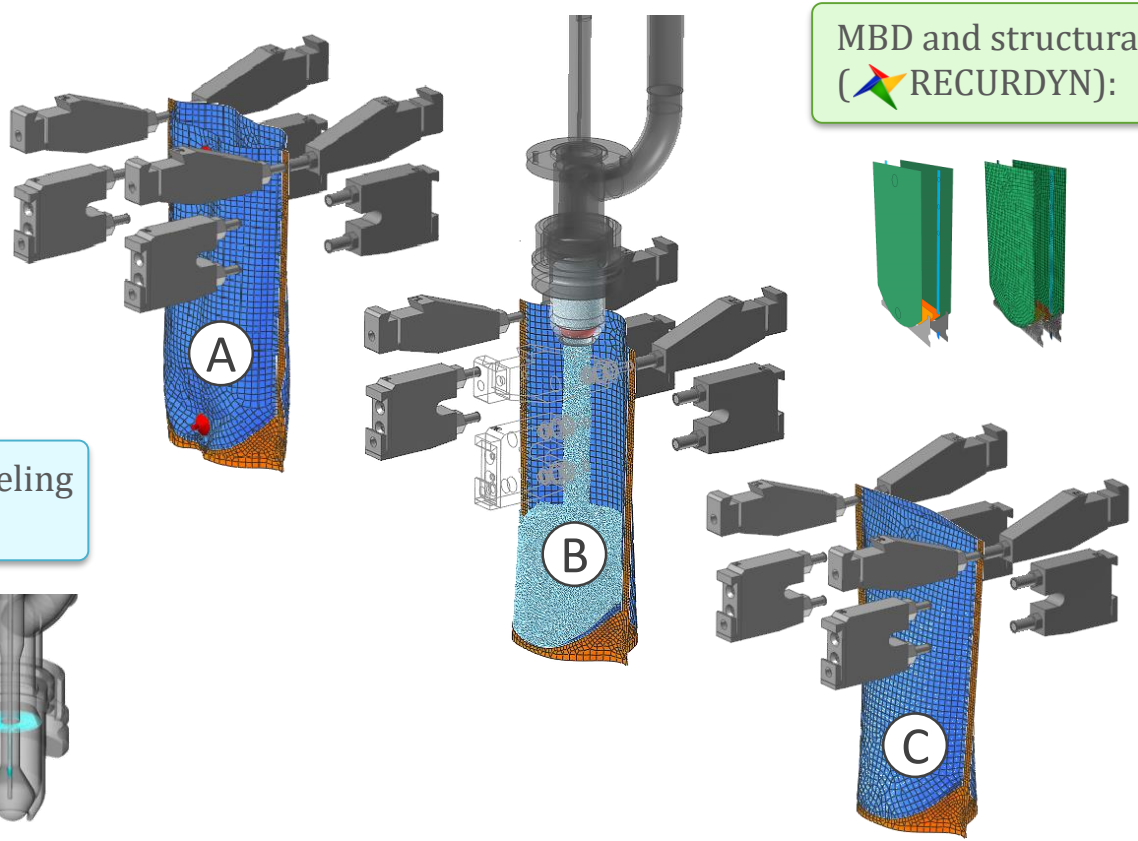
Large (nonlinear) displacements and deformations + multi-body → Multi-flexible-body dynamic simulation

Non-Newtonian free-surface fluid → Particle-based Computational fluid dynamic simulation

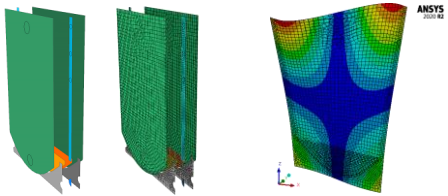
2-way fluid-solid interaction (FSI) → Efficient and robust coupling strategy

# Modeling

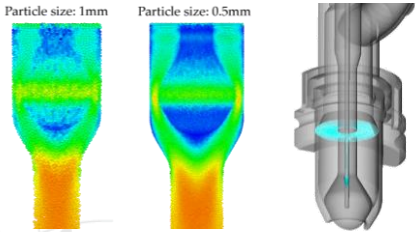
- Model setup




MBD and structural modeling (RECURDYN):

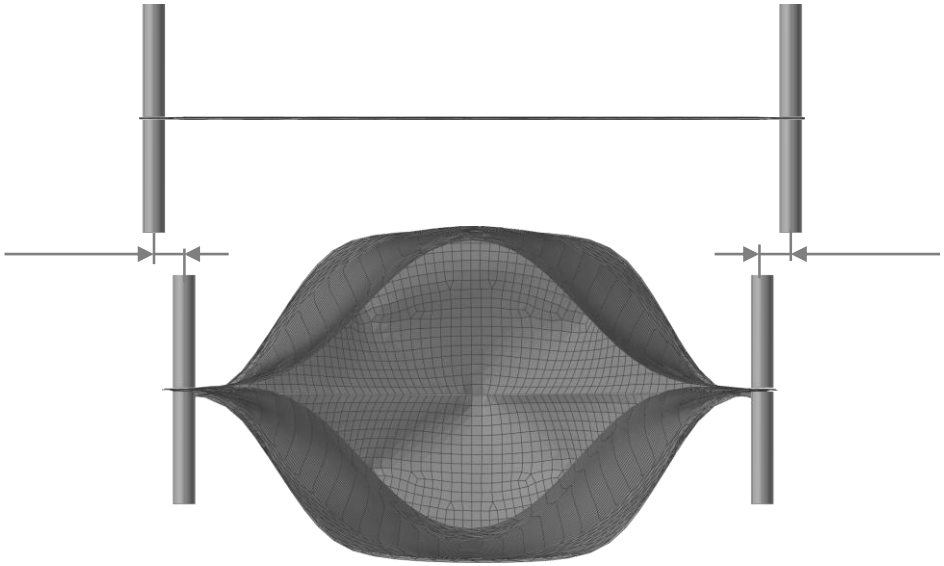



Filling and fluid modeling (Particleworks):



# Modeling

- Configurations
  - Opening span: configuration less (A1) or more (A2) open
  - Film material: less (M1) or more (M2) rigid film of the pouch



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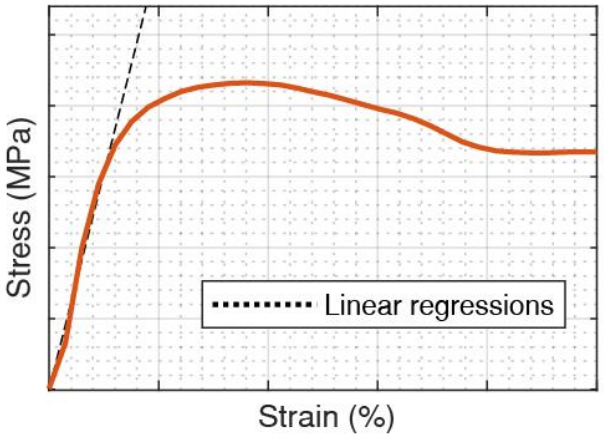
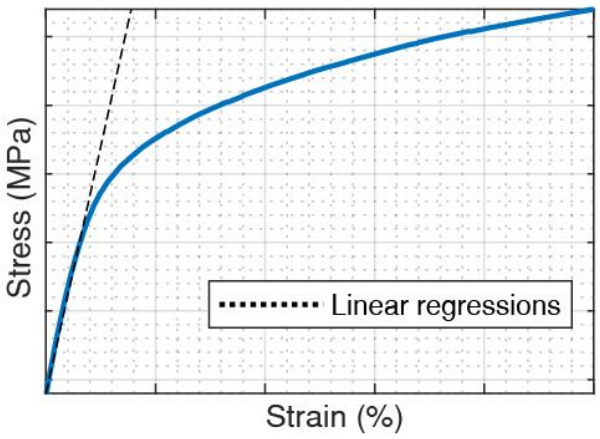



M1



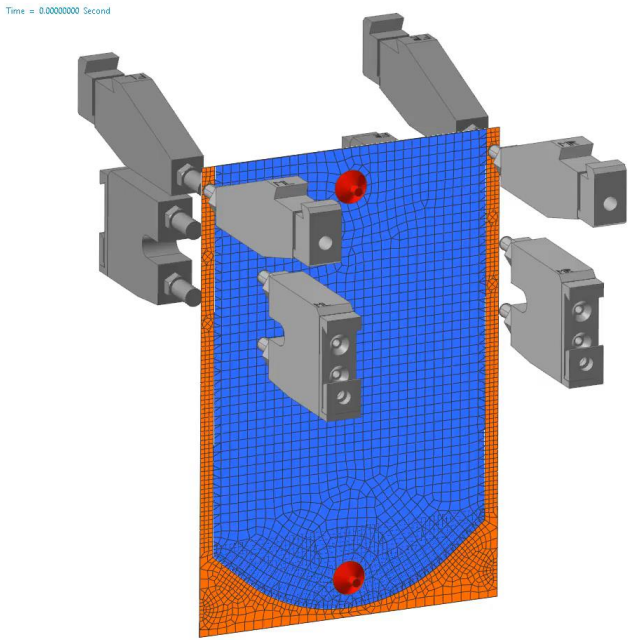

M2

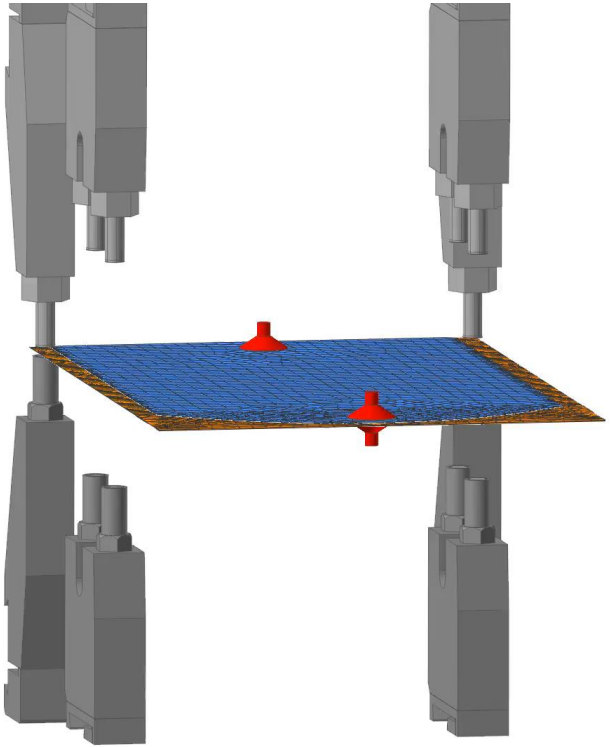
# Simulations and results

- Performed analyses:
  - **Full simulations** (i.e. opening, filling, and closing stages) with different configurations of the model (***A1***, ***A2***, ***M1***, ***M2***)
  - **DOE** varying the film properties (thickness and Young's modulus)
  - **Sloshing** analysis (filled pouch motion) considering ***A1***, ***A2*** and a set of machine speeds

# Simulations and results

- **Full simulations**



# Simulations and results

- **Full simulations**

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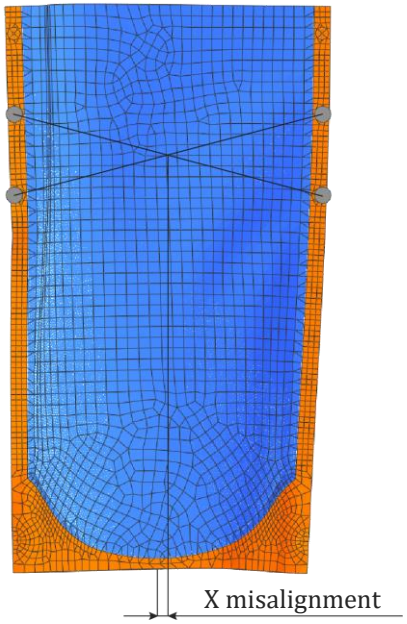
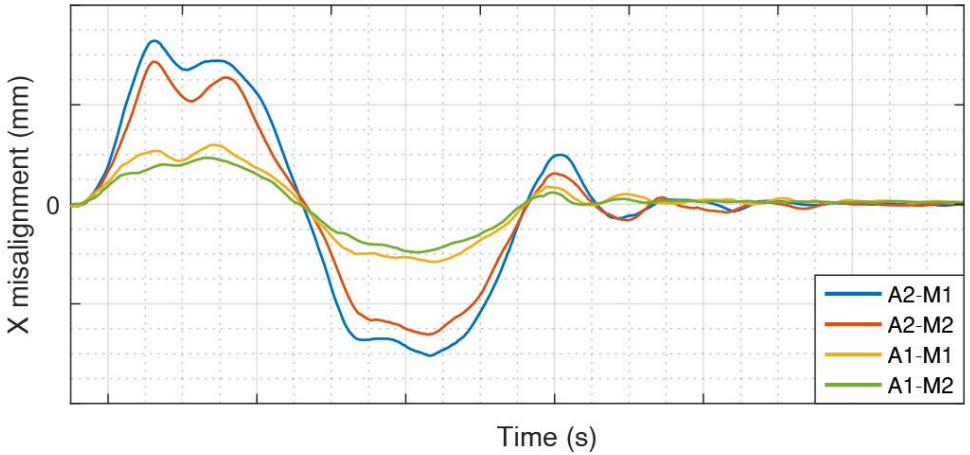
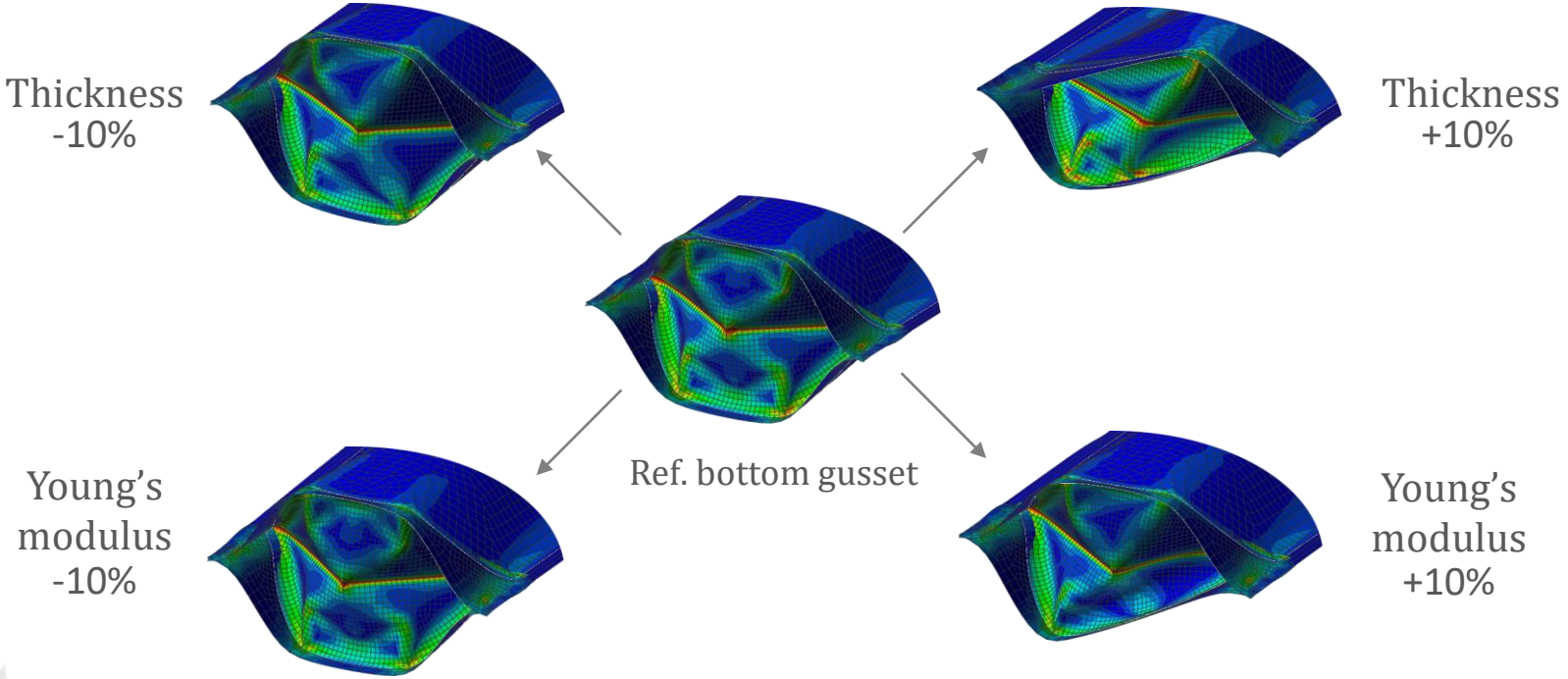



Lateral deformation (moving and closing)

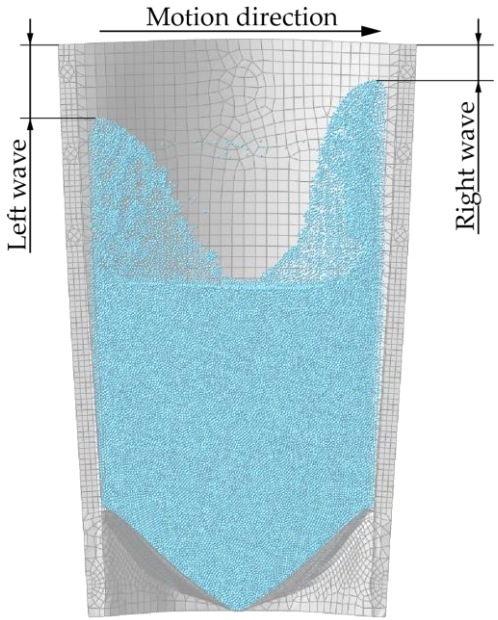
# Simulations and results

- **DOE**

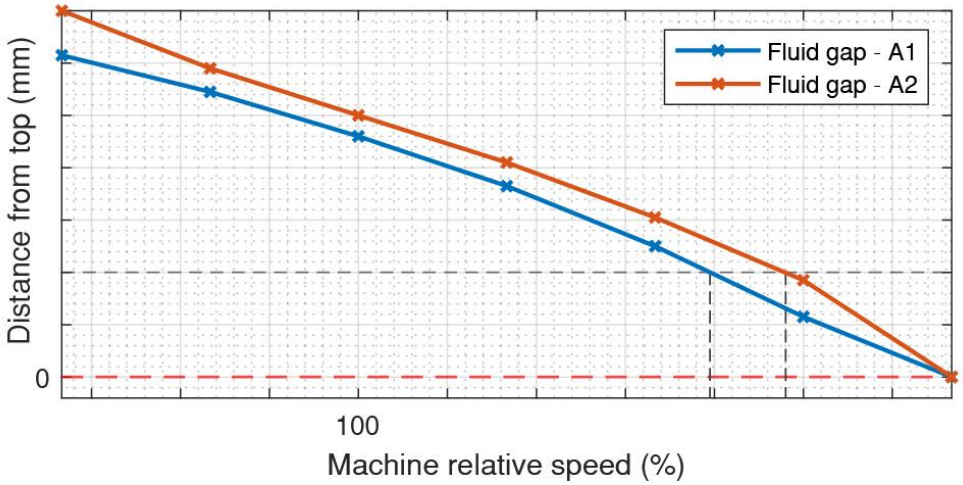


# Simulations and results

- **Sloshing**



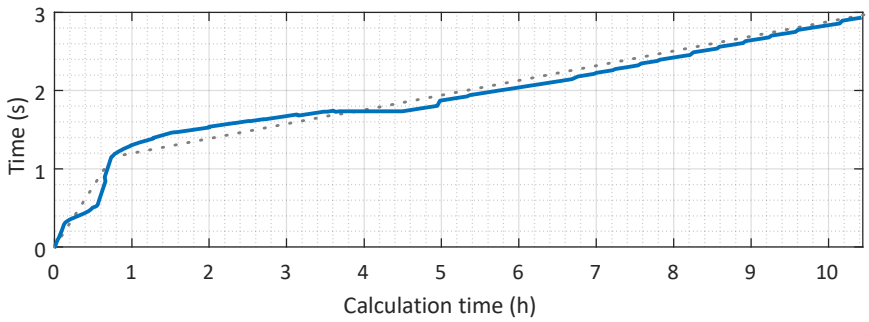

Internal sloshing



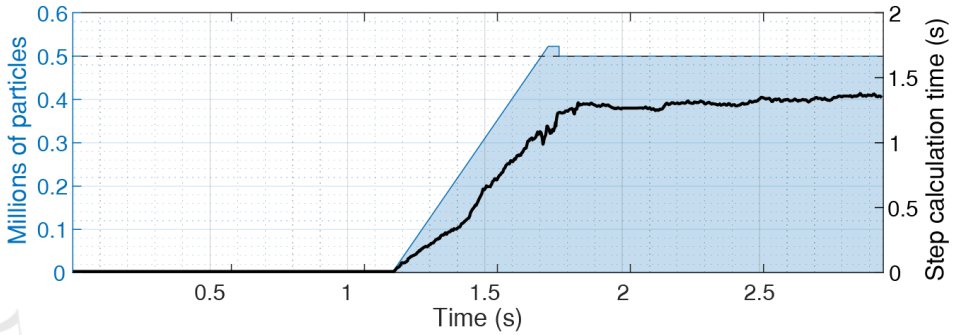

Minimum fluid-free gap due to the internal sloshing

# Simulations and results

- Calculation time




Calculation time in relation to the simulation time

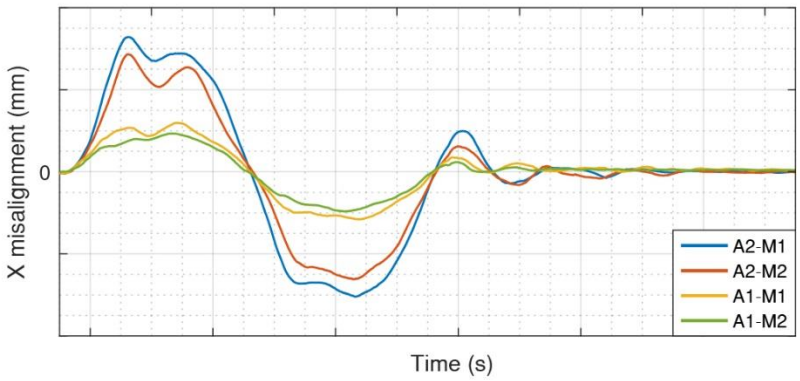



Number of particles and step calculation time during the simulation

# Conclusions

- Best combination of the parameters to
  - **minimize the deformation of the pouch through the stages**
  - maximize the machine speed under wave height constraints

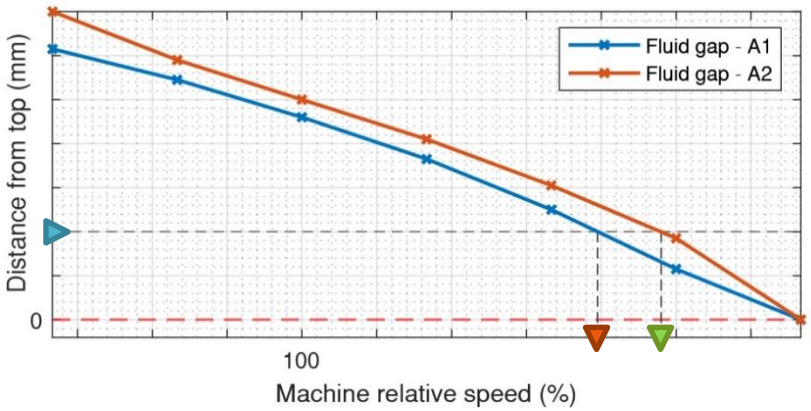
1. Select a configuration in which the pouch is less open
2. Select the material with the highest rigidity (Young's modulus)



# Conclusions

- Best combination of the parameters to
  - minimize the deformation of the pouch through the stages
  - **maximize the machine speed under wave height constraints**

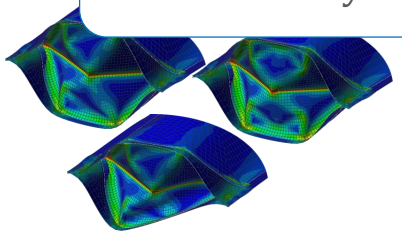
Select the most open configuration (A2)



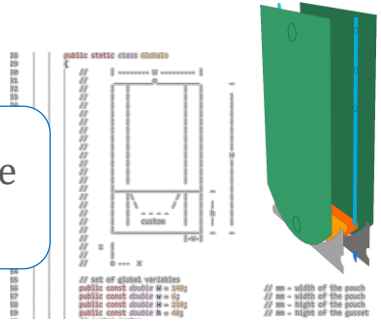
# Conclusions

- Capabilities and future work

Robustness of the model and sensitivity of the setup

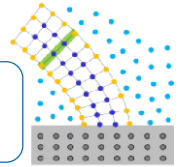


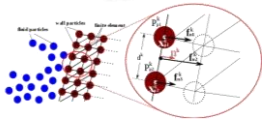
Suitability for a wide range of similar problems

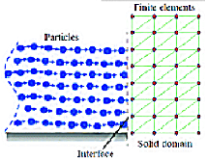


Calculation time

Novelty of the approach








*Zhang and Wan (2018)*

*Truong-Thi et al. (2018)*

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